

Face Recognition – Raspberry Pi 3B+ Project

This project contains a complete facial-recognition system built with Python, OpenCV, dlib, and the face_recognition library on a Raspberry Pi 3B+.

Files Included:

- headshots.py — captures training images
- train_model.py — generates encodings.pickle
- facial_req.py — performs live face recognition

How to Use:

1. Run headshots.py to collect dataset
2. Run train_model.py to train encodings
3. Run facial_req.py to perform recognition

Dependencies:

- face_recognition
- opencv-python
- numpy
- imutils
- dlib (requires cmake + boost)

Hardware:

- Raspberry Pi 3B+
- Pi Camera